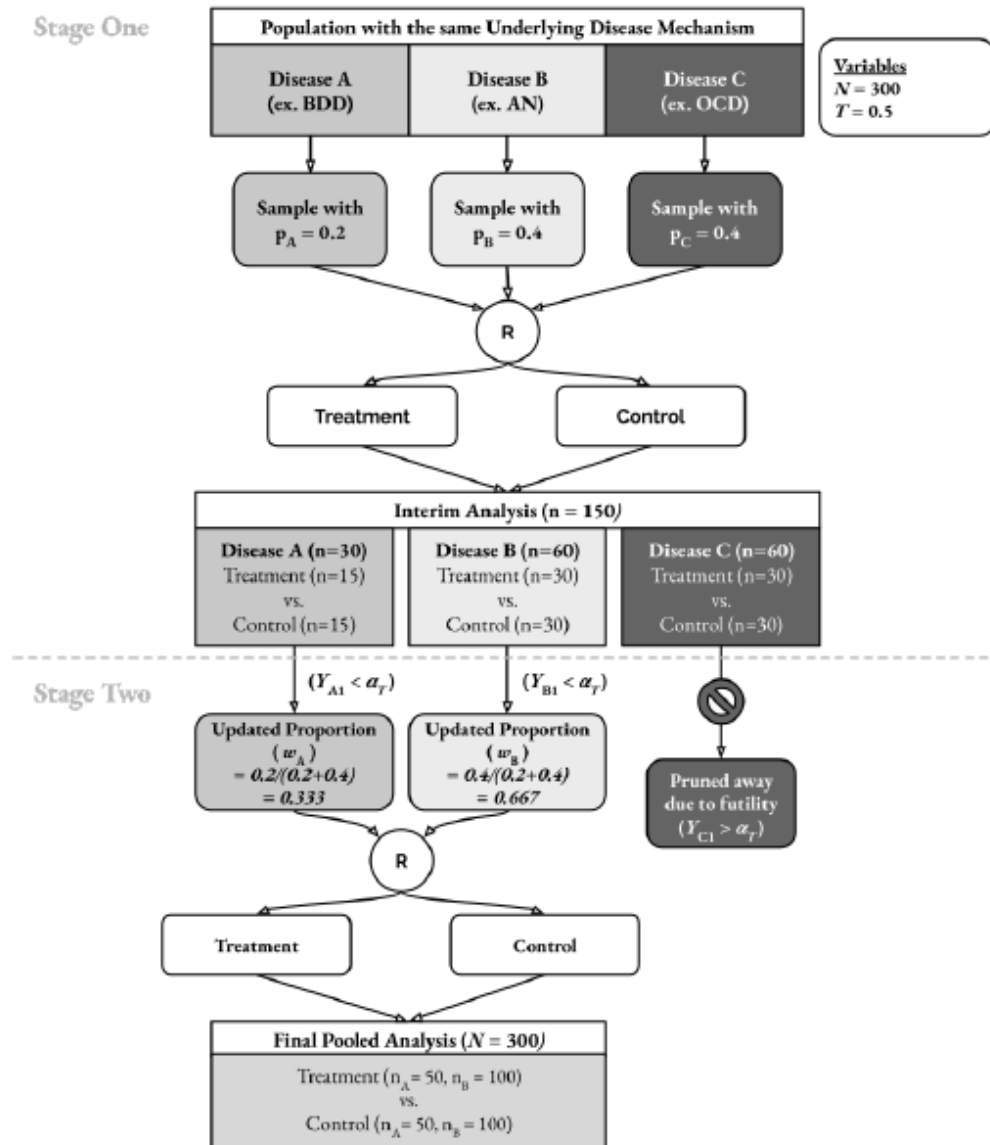


Standard Designs

RaBiT

SMARTIES

RaBiT Documentation



Above is a motivating example from Patel (2024). We obtain key metrics such as the total sample size for each basket, the true activity status of each basket, and the proportion of participants accrued during the interim analysis. To tailor the analysis to their specific trial designs, users provide additional inputs. This ensures maximum flexibility when computing test statistics. Refer to the screenshot below for guidance on transcribing the design above into inputs.

Inputs Panel

Vector Inputs (Multiple Values):

Total sample size of each basket := The number of participants per basket, using **only commas** to separate them. Ensure they are all positive even integers, so that after randomization, the cohort sizes in both the treatment and control arms are whole numbers. The length of the input vector corresponds to the number of baskets.

Relative ratio of each basket := The relative ratio of sample size in each basket, in the same order & format as above. Investigator can estimate effect sizes from published results.

Standardized effect size(Δ) of each basket := The expected effect size per basket, in the same order & format as above. Investigator can estimate this ratio via subtype prevalence from published results.

Accrual rate (individual/month) of each basket := number of individuals recruited per month in each basket, in the same order & format as the sample size input above.

Scalar Inputs(Single Value):

Nominal power := designated $1 - \beta$ level for the entire trial. One minus the type II error rate.

Proportion of participants accrued at interim analysis(t) := Interim analysis cutoff point, or % of total amount of participants recruited. This ratio should be consistent across all baskets.

Interim significance level(α_t) := designated α level for the interim analysis at proportion t . The basket with a p-value larger than this threshold will be dropped.

Trial-wide significance level(α) := designated α level for the entire trial, false positive rate overall.

Inputs:

What would you like to compute?

- ☒ Power
☐ Sample Size

Total sample size of each basket (N)

40,60,60,40

Standardized effect size (Δ) of each basket

0.3,0.3,0.5,0.2

Proportion of participants accrued at interim analysis (t)

0.3

Interim significance level (α_t)

0.5

Trial-wide significance level (α)

0.05

Accrual rate (individuals/month) of each basket

2,3,2,2

Calculate Now!

Inputs:

What would you like to compute?

- ☐ Power
☒ Sample Size

Nominal power

0.8

Relative ratio of each basket

2,3,3,2

Standardized effect size (Δ) of each basket

0.3,0.3,0.5,0.2

Proportion of participants accrued at interim analysis (t)

0.3

Interim significance level (α_t)

0.5

Trial-wide significance level (α)

0.05

Accrual rate (individuals/month) of each basket

2,3,2,2

Calculate Now!

Once all inputs have been filled as desired and vector inputs are ensured to be of the same length, click **Calculate Now!** to view outputs.

Outputs Panel

Current Design

An illustration produced to show the current design of the trial based on your inputs. To copy, simply right click>"copy image"or"save image as". If texts are not fitting well, please adjust the width of you browser until it fits properly.

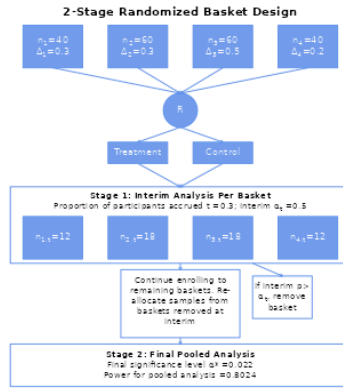
Design Summary

A test summary of current design of the trial based on your inputs. You can directly copy it since it is in text format.

Download as a Word Document

Click here to download all the content above, including text, images, and formatting, as a Word document.

Current Design



Design Summary: Estimating the minimum sample size required to achieve the nominal power

This trial uses a 2-stage, randomized basket design. A total of N=200 participants from 4 baskets are randomized 1:1 to the treatment and control arms, stratified by basket. The 4 baskets have N=(40,60,60,40) participants with standardized effect sizes $\Delta=(0.3,0.3,0.5,0.2)$, respectively.

The null hypothesis is that the standardized effect size Δ of the primary endpoint is zero ($H_0: \Delta=0$). The one-sided alternative hypothesis is that the standardized effect size of the primary endpoint is greater than zero ($H_a: \Delta>0$). Power calculations assume a basket to be truly active (under H_a) or truly non-active (under H_0).

Stage 1: Interim analysis within each basket

One pre-planned interim analysis will occur after a total of $0.3 \times 200 = 60$ participants have been enrolled. For each basket, a two-sample Z-test will compare the primary endpoint between arms. If the Z-test p-value is greater than the interim $\alpha=0.5$ threshold, accrual will be terminated for that basket. Accrual will continue for the remaining baskets. The planned number of participants from baskets removed during the interim analysis will be re-allocated to the remaining baskets so that the final analysis will maintain the original planned sample size of N=200 participants.

Stage 2: Final pooled analysis

After N=200 participants have been enrolled, a two-sample Z-test will compare the primary endpoint between arms pooling participants from the remaining baskets.

Power

The final pooled analysis has 80% power. To account for interim analyses, the final significance threshold $\alpha^*=0.022$ to maintain the trial-wide one-sided $\alpha=0.05$.

Trial Duration and Total Sample Size

The expected trial duration is 41.78 months (95% CI = 34.12 to 49.44 months). The total expected sample size is 215 participants (95% CI = 207 to 223 participants).

A Glimpse of the Statistical Method

For detailed derivation, please refer to Patel 2024 paper in the references.

Trial-wide Type 1 Error Rate:

$$\alpha = \sum_m Pr_{H_0}(V_m | \alpha^*, \alpha_t, m, \Delta) = \sum_m Pr_{H_0}(\cap_{i \in id(m)} Y_{i1} > Z_{1-\alpha_t}, V_m > Z_{1-\alpha^*}) \cdot (1 - \alpha_t)^{K-|id(m)|}$$

The overall type 1 error rate is a summed rejection probability under the null of different built, the rejection probability is assessed whether all baskets passing through interim and final stage.

Power

$$1 - \beta = \sum_m Pr_{H_{1g}}(V_m | \alpha^*, \alpha_t, m, \Delta)$$

Similarly, The power is a summed rejection probability under the alternative of different built.